



In the circuit shown above, all the resistors have the same value. If R_3 is removed from the circuit, how will the current I through the battery change?

- ☐ A. I will increase because there will be less resistance in the circuit. [28%]
- ☐ B. I will increase because there are fewer resistors in the circuit. [11%]
- ☐ C. I will not change because R_3 was in parallel with R_2 . [20%]
- ✓ ☒ D. I will decrease because there will be more resistance in the circuit. [38%]

Correct

38% answered correctly

Explanation:

Current in circuit decreases as total resistance increases

$I = \frac{V}{R}$

- V Voltage
- I Current
- R Total Resistance

Before

After

In an electric circuit, a voltage V generates a **current** I , which is the flow of charge per unit time in a conductor (eg, a wire, bulb, or resistor). Moreover, **Ohm's law** states that V is equal to the product of resistance R and current:

$$V = IR$$

In a circuit with multiple resistors in series and parallel, a total equivalent resistance for the entire circuit needs to be calculated. The total resistance of resistors in a **series circuit** R_s and in a **parallel circuit** R_p can be determined from the following combinations:

$$R_s = R_1 + R_2 + \dots + R_n$$

$$\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} + \dots + \frac{1}{R_n}$$

In this question, resistors R_2 and R_3 are in parallel and have the same resistance R_0 . Hence, the total equivalent resistance R_p for that specific branch is lower than just R_0 itself:

$$\frac{1}{R_p} = \frac{1}{R_0} + \frac{1}{R_0} = \frac{2}{R_0}$$

$$R_p = \frac{R_0}{2}$$

Hence, the total equivalent resistance will increase when removing R_3 . Because the branch containing only R_2 is in series with other resistors, the overall resistance of the circuit is increased.

According to Ohm's law, I is **inversely proportional** to R :

$$I = \frac{V}{R}$$

Therefore, an increase in overall circuit resistance for a constant voltage will result in less current provided by the battery.

(Choices A and C) R_2 and R_3 are in parallel. Hence, removing R_3 increases the resistance of the circuit and, consequently, I decreases.

(Choice B) Having fewer resistors in a circuit does not necessarily decrease resistance. For resistors in parallel, total resistance is less than the resistance of any of the single resistors and removing one resistor causes the total resistance to increase, not decrease.

Educational objective:

The total resistance increases when a resistor is removed from a parallel branch. Ohm's law can be used to find current, given resistance and voltage.